

Very Deep Convolutional Networks for Large-Scale Image Recognition (VGGNet)

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Introduction & Motivation

Background

The success of large-scale visual recognition has been driven by large public image repositories (ILSVRC) and high-performance computing systems (GPUs).

Since the victory of AlexNet in ILSVRC-2012, numerous attempts have been made to improve the original architecture.

Core Research Question

"How does the depth of a convolutional network affect its accuracy in the large-scale image recognition setting?"

Key Contributions

A thorough evaluation of networks with increasing depth using an architecture with very small (3x3) convolution filters.

ConvNet Configurations

Generic Design Principles

- Input:** 224 x 224 RGB images
- Filter Size:** 3 x 3 filters used consistently throughout the network
- Pooling:** 2 x 2 max-pooling (stride 2) applied after some convolutional layers
- Channel Width:** Starts at 64 in the first layer and increases by a factor of 2 after each max-pooling layer

Model Architecture

- Gradually increasing depth from 11 layers (Network A) to 19 layers (Network E)
- All configurations are followed by 3 Fully-Connected (FC) layers and a soft-max layer.

ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224 × 224 RGB image)					
conv3-64	conv3-64 LRN	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 conv1-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256 conv3-256
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

Network	A,A-LRN	B	C	D	E
Number of parameters	133	133	134	138	144

Number of Parameters by Configuration

Despite the large depth, the number of weights is not greater than a more shallow net with larger conv. layer widths and receptive fields.

Network	A / A-LRN	B	C	D	E
Parameters (millions)	133	133	134	138	144

Why 3 x 3 Filters?

Equivalent Receptive Field

- A stack of two 3x3 convolution layers = an effective receptive field of 5x5
- A stack of three 3x3 convolution layers = an effective receptive field of 7x7

Advantages of three 3x3 filters vs one 7x7 filter

More Discriminative: Incorporates three non-linear ReLU layers instead of a single one.

Fewer Parameters: 3x3 stack requires $27C^2$ weights vs $49C^2$ for a single 7x7 layer - an 81% reduction, acting as a regularizer.

1x1 Convolution & LRN Ablation

1x1 Convolution (Config C)

Adds non-linearity without changing the receptive field of conv. layers.

Config C underperforms Config D (both same depth, D uses 3x3 only) - shows that capturing spatial context matters more than adding non-linearity alone.

Local Response Normalisation (A vs A-LRN)

A-LRN (with LRN) shows virtually no improvement over A (without LRN): 29.7%/10.5% vs 29.6%/10.4% (top-1/top-5).

Conclusion: unlike AlexNet, LRN does not help on ILSVRC and only adds memory/compute cost - so LRN is dropped in deeper configs (B-E).

Training & Testing Framework

Training Strategy

- Mini-batch SGD (momentum 0.9, weight decay, dropout 0.5 in first two FC layers).
- **Weight init:** zero-mean Gaussian (variance 0.01), biases initialized to 0.
- **Pre-initialisation:** shallow Network A trained first; its weights initialize the first 4 conv layers and last 3 FC layers of deeper nets.
- **Learning rate:** starts at $1e-2$, divided by 10 on validation plateau (3 times total); training stopped after 370K iterations (74 epochs).
- **Scale Jittering range:** fixed $S=256$ or $S=384$, or randomly sampled S in $[256, 512]$ for multi-scale training.
- **Multi-GPU:** data parallelism across 4 GPUs gives 3.75x speedup vs single GPU; 2-3 weeks per net on 4 Titan Black GPUs.

Testing Strategy

- FC layers converted to conv layers (fully-convolutional), applied densely over the whole uncropped image at test scale Q .
- **Final score:** spatial sum-pooling of class score map; softmax of original + flipped image averaged.

Classification Experiments & Results

Config	Train Scale (S)	Test Scale (Q)	Top-1 Error	Top-5 Error
A	256	256	29.6%	10.4%
A-LRN	256	256	29.7%	10.5%
B	256	256	28.7%	9.9%
C	[256,512]	384	27.3%	8.8%
D	[256,512]	384	25.6%	8.1%
E	[256,512]	384	25.5%	8.0%

- Classification error consistently decreases as depth increases from 11 layers (A) to 19 layers (E).
- Scale jittering during training gives significantly better results than fixed-scale training.

ConvNet config. (Table I)	smallest image side		top-1 val. error (%)	top-5 val. error (%)
	train (S)	test (Q)		
A	256	256	29.6	10.4
A-LRN	256	256	29.7	10.5
B	256	256	28.7	9.9
C	256	256	28.1	9.4
	384	384	28.1	9.3
	[256;512]	384	27.3	8.8
	256	256	27.0	8.8
D	384	384	26.8	8.7
	[256;512]	384	25.6	8.1
	256	256	27.3	9.0
E	384	384	26.9	8.7
	[256;512]	384	25.5	8.0

ConvNet config. (Table I)	smallest image side		top-1 val. error (%)	top-5 val. error (%)
	train (S)	test (Q)		
B	256	224,256,288	28.2	9.6
C	256	224,256,288	27.7	9.2
	384	352,384,416	27.8	9.2
	[256; 512]	256,384,512	26.3	8.2
D	256	224,256,288	26.6	8.6
	384	352,384,416	26.5	8.6
	[256; 512]	256,384,512	24.8	7.5
	256	224,256,288	26.9	8.7
E	384	352,384,416	26.7	8.6
	[256; 512]	256,384,512	24.8	7.5

Comparison with State-of-the-Art

팀/모델	구성	Top-5 Error
GoogLeNet (1st place)	Ensemble	6.7%
VGG (2nd place)	7-net Ensemble	7.3%
VGG (improved)	2-net Ensemble	6.8%
VGG (single model)	Config D/E	7.0%

- In single-net performance, VGG (7.0%) outperforms a single GoogLeNet by 0.9%p.
- VGG won 1st place in the ILSVRC-2014 Localisation task.

Combined ConvNet models	Error		
	top-1 val	top-5 val	top-5 test
ILSVRC submission			
(D/256/224,256,288), (D/384/352,384,416), (D/[256;512]/256,384,512) (C/256/224,256,288), (C/384/352,384,416) (E/256/224,256,288), (E/384/352,384,416)	24.7	7.5	7.3
post-submission			
(D/[256;512]/256,384,512), (E/[256;512]/256,384,512), dense eval.	24.0	7.1	7.0
(D/[256;512]/256,384,512), (E/[256;512]/256,384,512), multi-crop	23.9	7.2	-
(D/[256;512]/256,384,512), (E/[256;512]/256,384,512), multi-crop & dense eval.	23.7	6.8	6.8

Conclusion

- The paper confirms that the "depth" of visual representations is highly beneficial for classification accuracy.
- State-of-the-art performance can be achieved using a conventional, simple ConvNet architecture by substantially increasing its depth with small (3x3) filters.

Thank You